

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA4305

COURSE OUTCOME COVERED

CO 5: Construct interoperable web services by leveraging JAX-RPC, WSDL, SOAP, and XML Schema for seamless data exchange across distributed systems. (L4, L5)

Assessment Tool: Viva Voce

Weightage: 10%

QUESTIONS:

Total Marks: 20

1. Define SOAP.

SOAP (Simple Object Access Protocol) is an XML-based protocol used for exchanging structured information between applications over a network.

2. Define WSDL.

WSDL (Web Services Description Language) is an XML-based language used to describe a web service, including its operations, messages, data types, and service endpoint.

3. What is an API?

API (Application Programming Interface) is a set of rules and methods that allows different software applications to communicate and exchange data.

4. Define XML Schema.

XML Schema (XSD) defines the structure, elements, attributes, and data types of an XML document and is used to validate XML data.

5. What is a Web Service?

A **Web Service** is a software component that allows different applications to communicate and exchange data over a network using standard web technologies.

6. Define REST.

REST (Representational State Transfer) is an architectural style for designing web services using HTTP methods such as GET, POST, PUT, and DELETE.

7. What is a Protocol?

A **protocol** is a set of rules that defines how data is transmitted and exchanged between devices or applications over a network.

8. Define Endpoint in web services.

An **endpoint** is the specific URL or network address through which a web service can be accessed.

Example:

<https://example.com/api/users>

9. What is a Service in web applications?

A **service** is a software component that performs a specific function and provides that functionality to other applications through an interface or API.

10. What is the difference between SOAP and REST?

SOAP	REST
Protocol	Architectural style
Mainly uses XML	Commonly uses JSON
More complex	Simpler
Generally heavier	Lightweight
Supports WS-Security	Commonly uses HTTPS, OAuth, JWT
Suitable for enterprise systems	Suitable for web and mobile applications

11. What is the role of WSDL in web services?

WSDL acts as a **contract or description** for a web service. It defines the available operations, input/output messages, data types, bindings, and service endpoint.

12. What is JSON and how is it used in APIs?

JSON (JavaScript Object Notation) is a lightweight data format used to represent and exchange structured data. REST APIs commonly use JSON to send requests and responses.

Example:

```
{  
  "name": "Rahul",  
  "age": 20  
}
```

13. What is HTTP and why is it used in web services?

HTTP (HyperText Transfer Protocol) is a communication protocol used to transfer data between clients and servers. Web services use HTTP to send requests and receive responses over the internet.

14. What is meant by request and response in web communication?

A **request** is a message sent by the client to the server asking for a resource or action.

A **response** is the message sent by the server back to the client containing the requested data or result.

Flow:

Client → Request → Server → Response → Client

15. What is the importance of XML in web services?

XML provides a **structured and platform-independent format** for representing and exchanging data. It is widely used in SOAP messages and WSDL documents.

16. What is the purpose of an API endpoint?

An **API endpoint** provides a specific URL through which a client can access a particular resource or operation of a web service.

Example:

`GET /api/products` → retrieves products.

17. What is the difference between HTTP and HTTPS?

HTTP transfers data without encryption.

HTTPS uses **TLS encryption** to protect data during transmission and also helps authenticate the server.

Therefore, HTTPS is more secure and is preferred for sensitive information.

18. What is meant by RESTful services?

RESTful services are web services designed according to REST principles. They use HTTP methods such as:

- **GET** – Retrieve data
- **POST** – Create data
- **PUT** – Update data
- **DELETE** – Delete data

19. What is a namespace in XML?

An **XML namespace** uniquely identifies XML elements and prevents naming conflicts when elements from different XML vocabularies are used together.

Example:

```
xmlns:h="http://example.com/hospital"
```

20. What is the role of a client and server in web services?

Client: Sends requests to access a web service or resource.

Server: Receives the request, processes it, and sends a response.

Example:

```
Client → HTTP Request → Server → Process → HTTP Response → Client
```

Code of Conduct:

I __P.Harika(192311243)__ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: P.Harika

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding	25	Demonstrates deep understanding of concepts with complete clarity	Explains concepts correctly with minor gaps	Partial understanding with errors or omissions	Unable to explain basic concepts
Application	25	Effectively applies concepts to new situations; solves problems logically	Applies concepts with minor errors	Limited ability to apply concepts; requires guidance	Unable to apply concepts effectively

Analytical Thinking	20	Critically analyzes scenarios with strong justification and reasoning	Provides reasonable analysis with some justification	Superficial analysis with weak reasoning	No analytical reasoning demonstrated
Communication	15	Communicates ideas clearly, confidently, and with appropriate terminology	Communicates clearly but with minor hesitation	Communication lacks clarity or structure	Unable to communicate ideas effectively
Response to Questions	15	Responds accurately and confidently, extending answers with depth	Responds correctly but with limited depth	Responds partially; requires prompting	Unable to respond to questions effectively